Do Now · retrieval
Hashing and applications

Name: _____ Class: _____ Date: _____

Answer four retrieval questions before reviewing earlier learning.

Q1

What is the encryption key in symmetric encryption?

Q2

What is the role of a public key?

Q3

Why encrypt a message in transit?

Q4

Compute 10110110_2 AND 00001111_2, then hexadecimal.

New Learning · key ideas

The important knowledge from today's lesson. Read, annotate, and explain it.

Core idea 1

Understand — A cryptographic hash transforms an arbitrary input into a fixed-size digest. It is designed to make reversing the digest infeasible, not impossible under all circumstances.

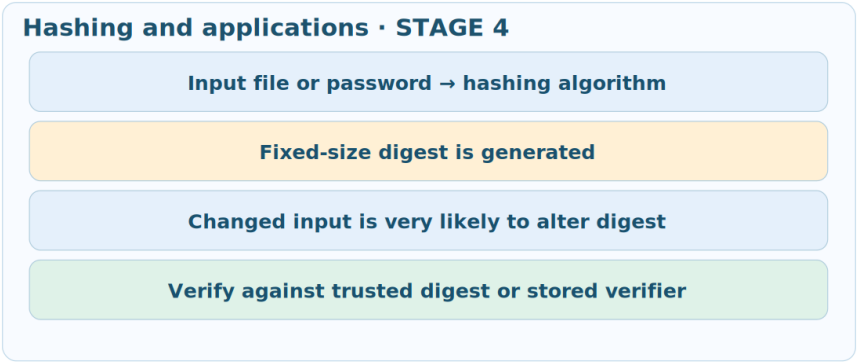
Core idea 2

Trace — A message digest supports integrity checking when the reference hash is trusted. Hash collisions can exist because outputs have finite length.

Core idea 3

Evaluate — Passwords should be processed using a salted, purpose-built slow password-hashing scheme, not simply a fast general-purpose hash and never reversible encryption.

Key visual · complete concept



Annotate the key visual

Explain which labels or features show the main process.

New Learning · applying the ideas

Vocabulary · use the correct technical terms

hash — Fixed-size digest of arbitrary input.

collision — Two inputs producing the same digest.

salt — Random value mixed into password hashing.

KDF — Key/password derivation function.

Check your understanding

Explain one key process in your own words using two of the terms above.

More animated examples are available in the original online lesson.

I Do · worked example

Problem

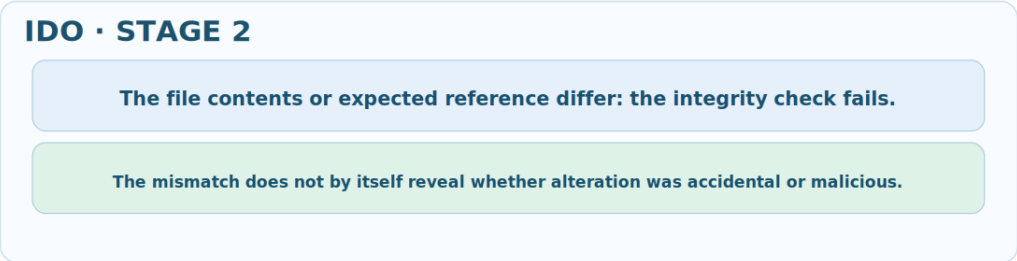
A downloaded file's trusted SHA-256 checksum differs from the computed checksum. What can you conclude?

Follow the worked method

1. Predict first — The file contents or expected reference differ: the integrity check fails.

2. Show method — The mismatch does not by itself reveal whether alteration was accidental or malicious.

Source model · final illustrated step



Why this method works

Identify the decision, rule or invariant that makes the method correct.

We Do · Try → Discuss → Improve

TRY independently in BLACK PEN; compare methods with a partner; improve your reasoning.

Guided problem · TRY

Predict first

Different random salts make identical passwords produce different stored verifiers.

DISCUSS

Compare how you reached the answer. Where could someone make an error?

IMPROVE

Add one precise correction or a stronger reason. Answers are not printed here.

You Do · independent practice

Answer these selected original questions in BLACK PEN. Other exam tasks remain online.

Question 1

Define cryptographic hashing and distinguish it from reversible encryption. [6]

Question 2

Explain how a trusted checksum helps verify a software download and give one limitation. [6]

Check that you have used the technical terms and explained your decisions.

You Do · extended practice

Answer these selected original questions in BLACK PEN. Other exam tasks remain online.

Question 3

Explain salted password hashing and why storing raw passwords or just fast unsalted hashes is weaker. [6]

Check that you have used the technical terms and explained your decisions.

Exit Ticket · reflect and retrieve

Exit Ticket

If a password is hashed, the server simply decrypts that hash during login.

Use key vocabulary from memory

1. Explain hash in your own words.

2. Explain collision in your own words.

3. Explain salt in your own words.

Confidence and next step

How confident are you? Thumbs up / side / down

Which question would you improve after checking the classroom model?

Check answers after the lesson using the teacher-guided source models.